



VERGE in VBS 2025

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Abstract. This paper presents VERGE, an interactive system designed for searching and browsing video content. VERGE allows users to explore collections of images extracted from videos, offering various search options like free-text queries, concept-based searches, color matching, detection of faces and people, as well as searches based on visual and semantic similarities. The system uses state-of-the-art AI for visual content analysis and can efficiently handle large datasets. VERGE also includes a web application that helps users form queries and displays the top search results, with the ability to combine different search methods. Improvements have been made to make the application more usable for non-expert users.

1 Introduction

VERGE is an interactive video retrieval system that enables users to search through collections of images extracted from videos using a web application. Users can perform queries, view the top-ranked results, and select and submit the content they are looking for. The system has been a consistent participant in the Video Browser Showdown (VBS) competition [13] since the competition began in 2012, and participated in the IVR4B¹ special session of the CBMI conference [1] for the first time in 2024. This year, we have extended the functionality of existing search methods. We also present an updated version of the redesigned web application, first introduced at IVR4B session, where it won first place. This new application enhances the user experience for novice users, addressing the complexity of the previous interface, which was designed mainly for experienced users. Compared to existing systems, VERGE stands out by combining many different retrieval methods, but in a way that maintains its user-friendly design and remains simple to use.

The structure of the paper is as follows: Sect. 2.1 outlines the framework of the system, while Sect. 2.2 presents the retrieval modalities that remained

¹ https://cbmi2024.org/?page_id=100#IVR4B.

unchanged. Section 3.1 describes the updated retrieval modalities, and Sect. 3.2 introduces the redesigned user interface and its new features. Finally, Sect. 4 discusses potential topics for future work.

2 The VERGE Framework

2.1 Overview of the Framework

Building on our previous work [15], we continue to use the four-layered architecture illustrated in Fig. 1. The first layer includes all the retrieval modalities; most of these process the datasets in advance, and their results are stored in a database. The second layer includes both the database and the Text to Video Matching service, which uses the corresponding modality to return results when a query is made. The third layer is the API with its different endpoints that fetch data from either the database or the services, providing the top results for each query. The final layer is the Web Application, which allows users to submit queries, communicates with the API, and displays the results to the user.

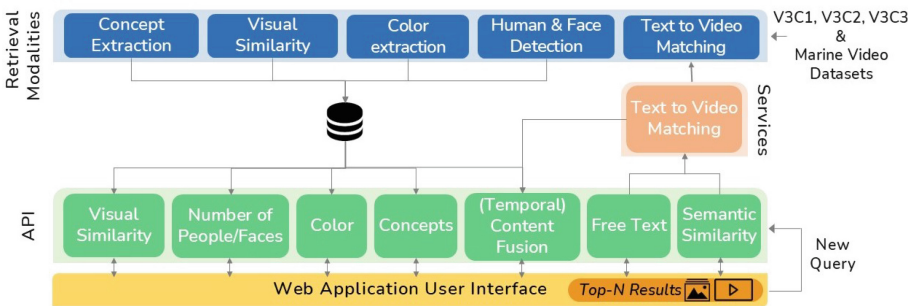


Fig. 1. The VERGE Framework

VERGE integrates several datasets, including V3C1 and V3C2 [17], the marine video dataset [21], and LapGyn100 (surgeries in laparoscopic gynecology), with the addition of V3C3 in this version. Details on the processing of these datasets can be found in [15].

2.2 Existing Retrieval Modalities

The following retrieval modalities have been part of VERGE since previous versions and continue to play a key role in the system's functionality. While they have not been modified in this version, they remain essential for helping users find video content effectively. Next, we outline these modalities, which are described in detail in [15].

Concept-Based Retrieval. This method labels keyframes using over 2000 concepts. It leverages pre-trained models like ViT [9] and EfficientNetV2 [20], trained on datasets such as ImageNet [18], Places365 [24], MS COCO [12], and Open Images V4 [10], as well as custom datasets like the TRECVID SIN task [14] and a sports dataset manually collected from the web. For more details, see Sect. 3.1 of [15].

Visual Similarity Search. To find visually similar images, DCNNs analyze content based on features from GoogleNet’s final pooling layer [16]. IVFADC indexing is used for efficient database vector storage, following [7].

Text to Video Matching Module. This module processes free-text queries to retrieve relevant video shots using an extension of the $T \times V$ network, the $T \times V + Objects$ [5]. $T \times V + Objects$ builds on $T \times V$ [4] by adding an object-based video encoder with transformers for finer visual granularity. Training utilizes four large-scale video caption datasets: MSR-VTT [23], TGIF [11], ActivityNet [3], and VateX [22].

Semantic Similarity Search. The video semantic similarity search, allows video shot to serve as a query. The system finds similar shots by comparing cross-modal embeddings from the Text to Video Matching Module module, offering comprehensive semantic comparisons beyond just visual aspects.

Human and Face Detection. This module is designed to determine the number of humans and human heads in each keyframe using YOLOv8 [8]. Pre-trained on the MS COCO dataset [12], the model is fine-tuned with the CrowdHuman dataset [19] to effectively handle partial occlusions common in crowded scenes.

Semantic Segmentation in Videos. This module generates pixel-level annotations of surgical tools and anatomical structures in each frame using LWANet [6], pre-trained on ImageNet [18] and fine-tuned on the EndoVis2017 [2] laparoscopic surgery dataset. Precise masks are extracted, enabling users to easily distinguish objects of interest against the surgical background and locate relevant scenes.

3 Recent Changes for VBS 2025

In preparation for the IVR4B special session of the CBMI 2024 conference, we updated some of the retrieval modalities and redesigned the user interface. Compared to the version presented at IVR4B, this updated version of the UI integrates additional datasets, such as LapGyn and Marine, and an analysis method specifically for the LapGyn dataset. Additionally, it includes some extra features that improve functionality.

3.1 Updated Retrieval Modalities

Late Fusion Approaches for Conceptual Text Matching. This year, we updated the two late fusion methods. One merges queries within the same shot, and the other across sequential video shots. The previous methods only combined free text with concepts, while the revised approaches allow any pair from free text, concepts, color, and people/faces. The algorithm for the late fusion remains the same in both cases, and is described in detail in [15].

3.2 Redesigned User Interface and Interaction Modes

The VERGE web application was completely redesigned (Fig. 2) to improve how users interact with the retrieval modalities that were described above. The previous version of the VERGE UI was basically designed for experienced users and had a complex menu layout. As new features were added to improve functionality, compromises were made in usability, resulting in a confusing interface that was difficult for novice users to navigate. In the new UI, while the results panel remains almost the same, the menu has been replaced with a clearer and more straightforward design, which supports two modes: advanced and novice user mode. A detailed description of the old UI is available in our previous work [15].

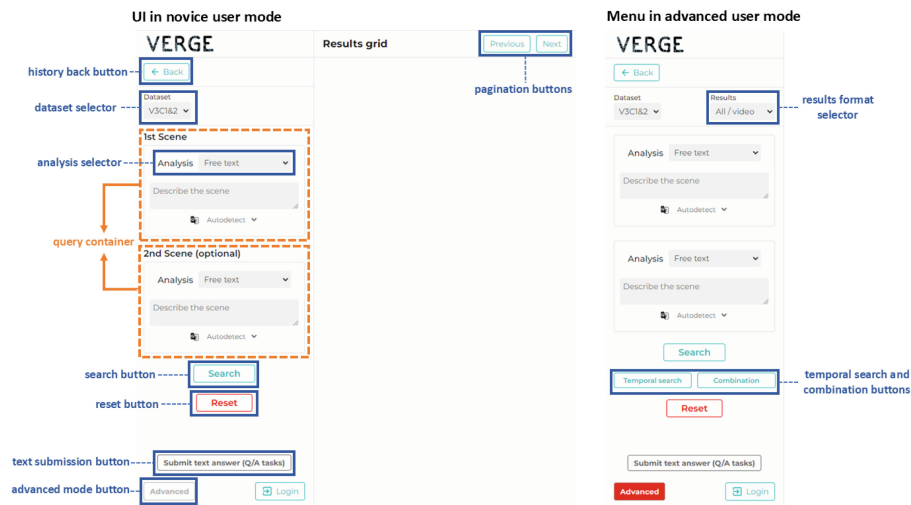


Fig. 2. The new VERGE User Interface

The new web application consists of two main parts: the menu and the results panel. By default, when users log in, they are in novice mode. At the top of the menu, there is the logo, a history back button, which is a new addition for VBS,

and a dataset selector. Below these elements, there is the main component of the menu: the query containers, which enable users to perform queries. Each container includes a title and a drop-down list with the available analyses, offering three options: free text, number of people, and number of faces. When an analysis is selected, its corresponding input field appears below the drop-down. The default option is the free text analysis, which now supports multiple languages by translating queries to English with the use of the Google Translation API², with an ‘Autodetect’ feature if the language is not explicitly selected. There are also different analysis options for each selected dataset. For the LapGyn dataset, users can draw rectangles to define areas of interest and then retrieve images of the dataset that contain tools in the selected areas. The input fields for the other analyses remain the same as in the old UI. Below the query containers, there are two buttons: ‘Search’ and ‘Reset’. The ‘Search’ button executes a regular search if only one query container has input, or a temporal search when both containers have input. The ‘Reset’ button restores the query containers to their default state, setting the analysis to free text and clearing any input fields. At the bottom of the menu, there are three buttons. The text submission button opens a modal where users can enter and submit their text responses for Q/A tasks. The ‘Login’ button allows users to connect to the application, which is required to submit results. Finally, the ‘Advanced’ button toggles between novice and advanced modes, providing access to additional features in the advanced mode. In the results panel, users can view the results of their queries. This component remains unchanged from the previous version, retaining the same styling and all of its functionalities. At the top right corner of the results panel, there are the pagination buttons that allow users to navigate through the different pages of the results.

In advanced mode, a results format selector appears next to the dataset selector. When ‘All/video’ is selected, all matching frames from the videos are returned, and when ‘First/video’ is selected only the first match from each video is returned. Additionally, the query containers offer two more analyses: concept-based text search, and color similarity, and the titles of the query containers are removed in this mode. The ‘Search’ button can only be used for a regular search by using one query container, and two new buttons are introduced between the ‘Search’ and ‘Reset’ buttons: ‘Temporal search’, which performs a temporal fusion of the two queries, and ‘Combination’, which returns the intersection of two queries.

To demonstrate the functionality of VERGE, we present four use cases. In novice user mode, a temporal search is performed using two free text analyses. In the ‘1st scene’ query container, the user selects ‘Free text’ and enters the description “a man playing piano in a forest”. In the ‘2nd scene’ container, the user again selects ‘Free text’ and enters the description “a band playing music by the sea”. After clicking the ‘Search’ button, the images that match the sequence of a man playing piano in a forest followed by a band performing by the sea, are displayed in the results panel. This use case can be seen in Fig. 3a. In the

² <https://cloud.google.com/translate>.

second use case, in advanced mode, a simple search query is performed using the color similarity analysis. The user selects specific colors for different areas of the scene: white for the upper left, blue for the middle, and black for the bottom right. After submitting the query, the system returns results where the specified color pattern is present, as it is depicted in Fig. 3b. Another example in novice mode is the following normal search, where the analysis ‘Number of people’ is selected. The user sets the value to 1, with a range of ± 1 , allowing the system to retrieve images that contain from 0 to 2 people in the scene. After clicking the ‘Search’ button, the results panel displays the frames where the number of people matches the specified range. In the final use case, in advanced mode, the ‘Concepts’ analysis is used. The user selects ‘Concepts’ from the drop-down and then “ice skating” from the list of the concepts that appears. As before, the system retrieves images that depict the activity of ice skating, and displays them in the results panel.

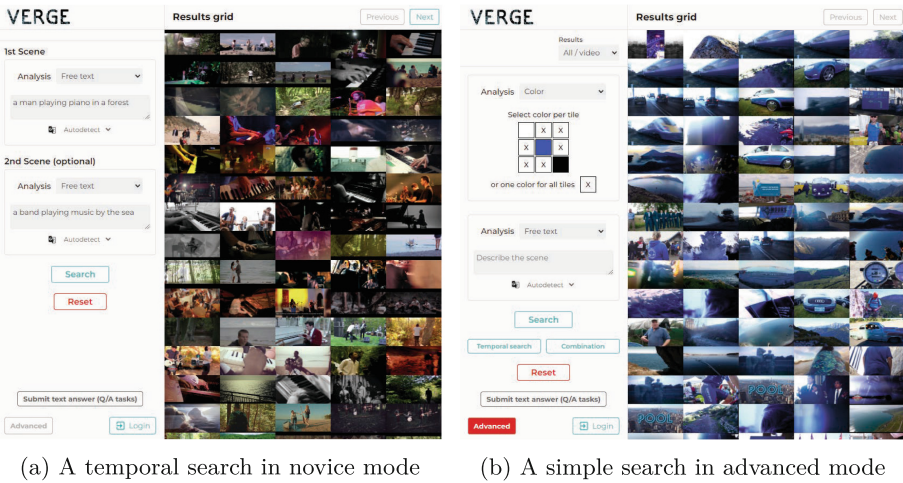


Fig. 3. Two use cases of the VERGE UI functionality

4 Future Work

In the next phase of research and development, we will focus on the keyframe extraction process. Our goal is to replace the predefined shots with the use of advanced algorithms to dynamically identify keyframes. This will lead to more accurate segmentation of the video content and better retrieval results. Additionally, we plan to further improve the web application, with a focus on its performance with large videos, which will provide a smoother experience for the users.

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